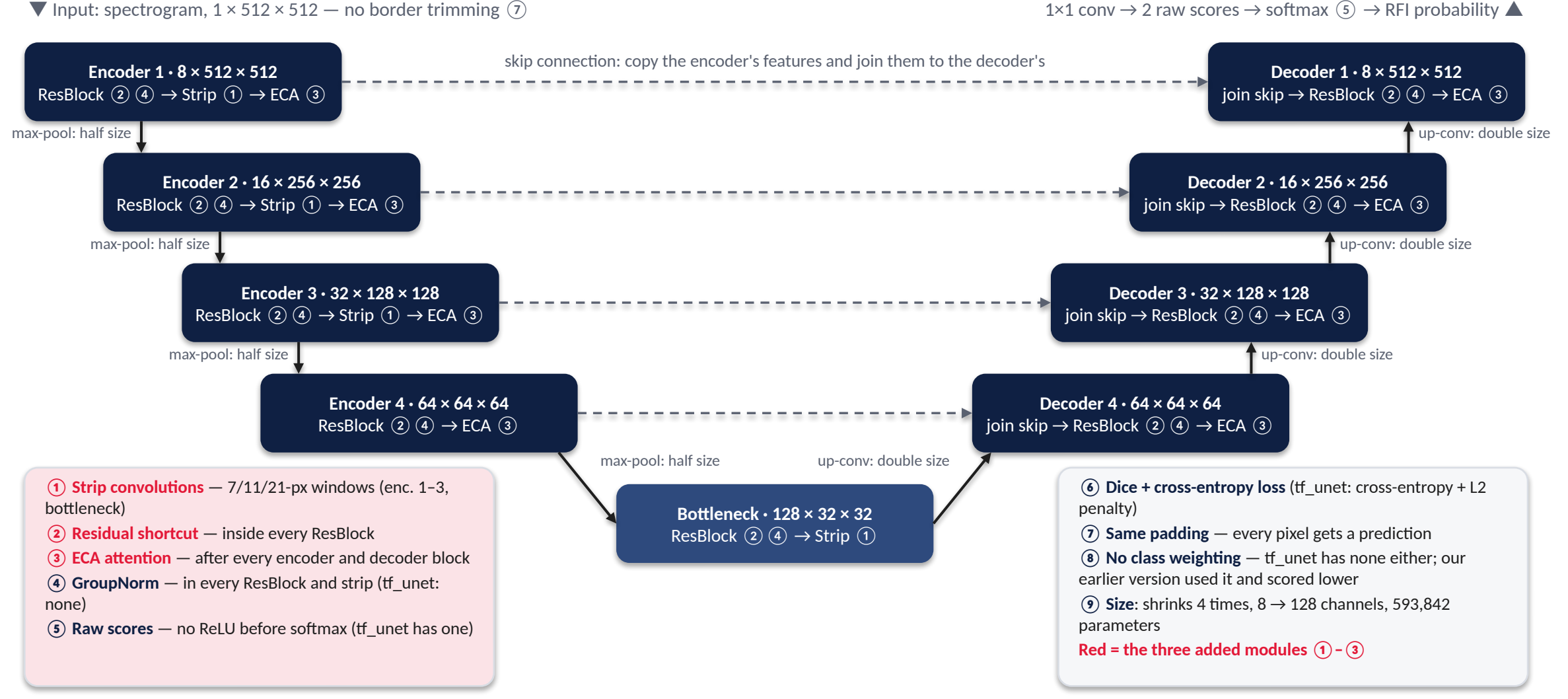

▼ Input: spectrogram, 1 × 512 × 512 — no border trimming ⑦
1×1 conv → 2 raw scores → softmax ⑤ → RFI probability ▲
Encoder 1 · 8 × 512 × 512
ResBlock ② ④ → Strip ① → ECA ③
skip connection: copy the encoder's features and join them to the decoder's
Decoder 1 · 8 × 512 × 512
join skip → ResBlock ② ④ → ECA ③
max-pool: half size
up-conv: double size
Encoder 2 · 16 × 256 × 256
ResBlock ② ④ → Strip ① → ECA ③
Decoder 2 · 16 × 256 × 256
join skip → ResBlock ② ④ → ECA ③
max-pool: half size
up-conv: double size
Encoder 3 · 32 × 128 × 128
ResBlock ② ④ → Strip ① → ECA ③
Decoder 3 · 32 × 128 × 128
join skip → ResBlock ② ④ → ECA ③
max-pool: half size
up-conv: double size
Encoder 4 · 64 × 64 × 64
ResBlock ② ④ → ECA ③
Decoder 4 · 64 × 64 × 64
join skip → ResBlock ② ④ → ECA ③
max-pool: half size
up-conv: double size
Bottleneck · 128 × 32 × 32
ResBlock ② ④ → Strip ①
① Strip convolutions — 7/11/21-px windows (enc. 1–3, bottleneck)
② Residual shortcut — inside every ResBlock
③ ECA attention — after every encoder and decoder block
④ GroupNorm — in every ResBlock and strip (tf_unet: none)
⑤ Raw scores — no ReLU before softmax (tf_unet has one)
⑥ Dice + cross-entropy loss (tf_unet: cross-entropy + L2 penalty)
⑦ Same padding — every pixel gets a prediction
⑧ No class weighting — tf_unet has none either; our earlier version used it and scored lower
⑨ Size: shrinks 4 times, 8 → 128 channels, 593,842 parameters
Red = the three added modules ① – ③